

Delphi

Evaluation & validation of the agent swarm

SANKAR SUBBAYYA · GOOGLE I/O HACKATHON · 23 MAY 2026

What needs validating

A swarm of N Gemini 3.5 Flash sub-agents produces a forecast. Three layers of trust must hold for the output to be defensible.

- **Each agent reasons in character.** Distinct personas – not regression to a centrist mean.
- **The swarm aggregates honestly.** Math is right; per-agent failures are surfaced, not silently dropped.
- **End-to-end runs are reproducible.** Same inputs, same shape of output, same shock-response behavior.

This deck reports on what we have measured – and what we have not.

Four layers of validation

LAYER	WHAT IT PROVES
Live runs	Real Gemini, real grounding, real demographic spread. End-to-end on stage.
HTTP & WebSocket	Request lifecycle, stream message ordering, shock endpoint, drill-down.
Integration harness	Full pipeline against a <i>FakeClient</i> – runs offline, no API key needed.
Unit	Aggregator math, persona samplers, JSON extraction, mode-specific schemas.

26 AUTOMATED TESTS • 0.82-SECOND SUITE RUNTIME

Layers 1–2 • automated tests

MODULE	TESTS	COVERAGE
Aggregator	5 / 5	mean, $\pm 1\sigma$, distribution buckets, demographic grouping
Personas	3 / 3	weighted samplers, axis invariants, distribution weights
Agent JSON	3 / 3	markdown fences, garbage payloads, empty responses
Integration	7 / 7	persona-gen → swarm → aggregate → shock, with FakeClient
HTTP / WebSocket	8 / 8	create → stream → drill-down → shock → 404 / 409 paths

ALL 26 PASSING • ZERO FLAKES ACROSS 5 CONSECUTIVE DRESS-REHEARSAL RUNS

Layer 3 • end-to-end through the HTTP boundary

The integration harness exercises the same code path a real client takes – through FastAPI, through the WebSocket stream, through the background task scheduler.

- **POST /swarm/run** creates `RunState`, schedules `asyncio.create_task`
- **WS /.../{id}/stream** emits `personas[]` → `response × N` → `done + forecast + summary`
- **GET /.../persona/{pid}** loads full reasoning trace from in-memory store
- **POST /.../shock** appends shock to question, re-runs swarm, returns new forecast + new summary

Verified offline: state-machine correctness, WS message ordering, per-agent timeout (25 s) handling, CORS, 404 / 409 edge cases.

Live run • convergent question

QUESTION

Will rising gas prices affect Walmart's performance?

CONFIGURATION

N = 20 • forecast mode • live grounding

HEADLINE

91.8%

±1σ band [87%, 96%]

20 / 20 personas reasoned • 1 timed out (graceful)

DISTRIBUTION

0.0 – 0.2	0 %
0.2 – 0.4	0 %
0.4 – 0.6	0 %
0.6 – 0.8	0 %
0.8 – 1.0	100 %

INTERPRETATION

Plumber, dental hygienist, logistics manager reached the same conclusion *from different demographic angles*. Variance was in **reasoning**, not in position – the expected shape on a question with one defensible answer.

Live run · shock-responsive question

QUESTION

Will the Fed cut interest rates in Q3 2026?

CONFIGURATION

N = 5 · forecast mode · live grounding

	BEFORE	AFTER
Headline	17.0%	81.2%
±1σ band	[12.5, 21.5]	[76.5, 86.0]
Failed	0	1

SHOCK INJECTED

"Surprise May CPI prints 2.1%, well below forecasts."

+64.2 pp

Δ HEADLINE IN < 15 SECONDS

The same personas with the same demographic vectors re-reasoned against the new context – not a string substitution. The new summary paragraph cited the inflation print, not the question keywords.

Reasoning quality · three voices, one question

QUESTION · WILL APPLE SHIP GLASSES IN 2027?

"I keep hearing on the podcasts I listen to between jobs that Apple is feeling the heat from Meta and Google... personally I think it's a waste of money when rent and groceries are through the roof."

— 24YO APPRENTICE PLUMBER, OREGON · POSITION 80%

"Apple is desperate to lock us into the next surveillance-capitalism frontier... I really hope they flop — most of us in Chicago are struggling to pay rent, not looking to drop hundreds on a glorified face-spy tool."

— 22YO SELF-TAUGHT IT SPECIALIST, CHICAGO · POSITION 85%

"From a supply-chain standpoint, shipping a basic accessory is highly doable, though I'm naturally skeptical of Silicon Valley's optimistic timelines."

— 51YO LOGISTICS MANAGER, OHIO · POSITION 75%

Showcase · forecast mode · consumer pricing

QUESTION

Will most Americans pay \$200/month for a personal AI assistant by 2027?

N 8 MODE forecast

HEADLINE

1.75%

±1σ band [0.4%, 3.1%]

POPULATION SYNTHESIS

American consumers overwhelmingly reject the prospect of a \$200 monthly AI subscription by 2027, viewing the price as an exorbitant tech-bubble luxury that conflicts with basic household expenses.

REASONS AGAINST

- Severe competition from free bundled software
- Prioritisation of rising grocery & utility costs
- Widespread perception of poor value-for-price

Two hundred dollars a month is a utility bill or a week of groceries, and there is no way average working folks are going to hand that over

Showcase · pretest mode · brand tagline

TAGLINE TESTED

Walmart – your community's everyday partner.

N 8 MODE pretest

HEADLINE

negative

consensus, with one outlier in support

POPULATION SYNTHESIS

The tagline faces strong negative consensus – respondents widely reject the "community partner" framing as cynical corporate marketing that contradicts Walmart's history of displacing local businesses.

REASONS AGAINST

- Displacement of local businesses
- Substandard worker compensation
- Environmental & ecological degradation

If they want to partner with our neighborhood block associations to keep doing good, I'd welcome them with open arms.

— 68YO RETIRED POSTAL WORKER, ATLANTA GA (THE LONE POSITIVE VOTE)

Showcase · stress-test mode · pricing defence

DECISION TESTED

Spotify is raising prices 5% annually, indexed to AI training costs, starting July 2026.

N 8 MODE stress-test

HEADLINE

objects

unanimous across all demographics

POPULATION SYNTHESIS

Consumers universally reject the move, viewing the indexing of subscription fees to corporate AI training costs as an unfair tax on everyday users.

OBJECTIONS RAISED

- Subsidising corporate AI research
- Compounding subscription fatigue
- Devaluation of human musical craftsmanship

If I tried to pass off my internal software costs as a mandatory annual rate increase, my clients would rightly take their pets elsewhere.

— 65YO VETERINARY CLINIC OWNER, GRAND RAPIDS MI

What we have *not* yet validated

- **Calibration against real polling baselines.** Synthetic forecasts have not been benchmarked against Polymarket, Pew, Gallup, or Good Judgment Project on matched questions.
- **Demographic representativeness.** Distributions are coarse US-adult proxies; not Census-aligned, no region-level granularity below seven buckets.
- **Behavior at $N = 1000$.** Framework supports it. Only $N \leq 20$ demonstrated live within hackathon time / API quota.
- **Cross-model drift.** Locked to gemini-3.5-flash for this evaluation. No comparison runs against gemini-3.5-pro or earlier Flash variants yet.
- **Adversarial persona stability.** Whether personas hold character under deliberately ambiguous or politically charged prompts.

HONEST READ · POSITIONS ARE PLAUSIBLE AND SHOCK-RESPONSIVE, BUT UNCALIBRATED. THIS IS A PROTOTYPE OF A PRIMITIVE, NOT A FORECASTING PRODUCT YET.

Calibration roadmap

- **Backtest harness** known historical events (Fed decisions, election outcomes, product launches) → query Delphi as if asking *before* the event → score Brier loss vs. Good Judgment Project median
- **Polling-matched calibration** matched questions from Pew & Gallup; measure mean shift, distribution overlap, KS statistic
- **Persona-quality eval** blind-review N=200 traces for diversity, in-character consistency, sources-cited rate, reasoning length stability
- **Stress harness** per-agent timeout rate · summary-generation failure rate · rate-limit behaviour at N=1000 · cost per run

VALIDATION IS THE MOAT. THE INTERESTING RESEARCH STARTS AFTER THIS DECK.

Delphi

SYNTHETIC POPULATIONS AS A COMPUTATIONAL SUBSTRATE

26 automated tests · multiple live runs · honest disclosure
of what is and is not yet calibrated.

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